

House Price Prediction using Linear Regression

Artificial Intelligence & Machine Learning Project Report

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed. One of the most widely used supervised learning algorithms is Linear Regression, which is used for predicting continuous numerical values.

In this project, a Linear Regression model was developed to predict house prices using the California Housing Dataset available in the scikit-learn library. The project demonstrates the complete Machine Learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and visualization.

The main objective of this project is to understand how regression algorithms work and how machine learning models can be used for real-world prediction problems such as house price estimation.

2. Objective of the Project

The major objectives of this project are:

- To understand the Machine Learning workflow.
 - To perform Exploratory Data Analysis (EDA).
 - To preprocess and analyze housing data.
 - To build a Linear Regression model.
 - To evaluate the model using different performance metrics.
 - To visualize actual and predicted house prices.
 - To understand the strengths and limitations of Linear Regression.
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3. Dataset Description

The California Housing Dataset was used in this project. The dataset is available in the scikit-learn library and contains information collected from California districts.

The dataset contains several numerical features related to housing and population statistics. The target variable is the median house value.

Features of Dataset

Feature Name	Description
MedInc	Median income in the area
HouseAge	Average house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Total population
AveOccup	Average occupancy
Latitude	Latitude location
Longitude	Longitude location

Target Variable

- **MedHouseVal** → Median House Value

The dataset is clean and suitable for regression tasks.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is an important step in Machine Learning. It helps in understanding the structure and distribution of data before training the model.

The following EDA operations were performed:

- Displayed first five rows using `head()`
- Checked dataset information using `info()`
- Checked statistical summary using `describe()`
- Checked missing values using `isnull().sum()`
- Generated correlation heatmap

Observations from EDA

- The dataset did not contain significant missing values.
- Median income showed strong positive correlation with house prices.
- Longitude and latitude also affected house prices.
- Some features had weak correlations with the target variable.
- Most features were numerical, making the dataset suitable for Linear Regression.

A correlation heatmap was plotted to visualize relationships among variables.

5. Data Preprocessing

Data preprocessing is the process of preparing raw data before training the model.

The following preprocessing steps were performed:

1. Feature selection
2. Separating input and target variables
3. Splitting dataset into training and testing sets

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

The `train_test_split()` function from scikit-learn was used for splitting the dataset.

6. Model Building

A Linear Regression model was built using the `LinearRegression()` class from scikit-learn.

Linear Regression attempts to establish a linear relationship between independent variables and the dependent variable.

The model was trained using:

- Training features (`x_train`)
- Training labels (`y_train`)

After training, predictions were generated on the test dataset using:

```
model.predict(X_test)
```

The predicted values were compared with actual house prices.

7. Model Evaluation

The model performance was evaluated using the following metrics:

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

Lower MAE indicates better model performance.

Root Mean Squared Error (RMSE)

RMSE measures the square root of average squared prediction errors.

Lower RMSE indicates better prediction accuracy.

R² Score

R² score measures how well the model explains variance in the data.

- R² close to 1 → Good model
- R² close to 0 → Poor model

Evaluation Results

The model produced reasonable prediction accuracy on the California Housing dataset.

The scatter plot between actual and predicted values showed that many predictions were close to actual values.

8. Visualization

Data visualization helps in understanding model performance more effectively.

The following visualizations were created:

- Correlation Heatmap
- Actual vs Predicted Scatter Plot

Actual vs Predicted Plot

The scatter plot compared actual house prices with predicted house prices.

A red reference line was plotted to indicate ideal predictions. Points near the line represented accurate predictions.

The visualization showed that the Linear Regression model was able to capture general trends in the housing data.

9. Advantages of Linear Regression

- Simple and easy to understand
 - Fast training process
 - Works well for linear relationships
 - Easy interpretation of results
 - Useful baseline model for regression tasks
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10. Limitations of Linear Regression

- Assumes linear relationship between variables
 - Sensitive to outliers
 - Performance decreases for complex datasets
 - Cannot capture nonlinear relationships effectively
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11. Improvement Ideas

The following improvements can enhance model performance:

- Apply feature scaling
- Remove outliers
- Use feature engineering techniques
- Perform hyperparameter tuning
- Use advanced algorithms such as:
 - Random Forest Regression
 - Decision Tree Regression
 - XGBoost
 - Gradient Boosting

Cross-validation can also improve model reliability.

12. Conclusion

In this project, a House Price Prediction model was successfully developed using Linear Regression and the California Housing Dataset.

The project demonstrated the complete Machine Learning lifecycle including:

- Data loading
- Exploratory Data Analysis
- Data preprocessing
- Model training
- Prediction
- Evaluation
- Visualization

The Linear Regression model achieved reasonable prediction accuracy and provided valuable insights into housing price trends.

This project helped in understanding the fundamentals of Machine Learning, regression analysis, and model evaluation techniques.